

MD MAHIR JAWAD

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PROFESSIONAL EXPERIENCE

Machine Learning Engineer

August 2025 – Present

Penta Global Ltd (Full-Time)

Dhaka, Bangladesh

- Spearhead the BIM Model Automation Project, engineering automated pipelines to translate Scaffolding, PDF, and CAD inputs into dynamic 3D BIM models.
- Developing "Plinth", an AI-based BIM web application, integrating BIM Automation that includes 3D Modelling from 2D, QTO, Model Audit: A All-in-one BIM platform.

Machine Learning Engineer Intern

May 2025 – July 2025

Penta Global Ltd

Dhaka, Bangladesh

- Designed a programmatic machine learning workflow to fine-tune a Segment Anything Model 2 (SAM 2) LORA adapter and train a YOLO locator using annotated floorplan images for accurate area calculations.
- Developed multiple end-to-end projects demonstrating comprehensive expertise in Large Language Models (LLMs) and MCP.

UI/UX Developer Intern

July 2024 – October 2024

Synergy Bangladesh

Dhaka, Bangladesh

- Designed and developed the canvas-bd.com web platform utilizing Odoo, seamlessly integrating modern e-commerce web design with backend ERP systems.

EDUCATION

Masters of Science in Computer Science

June 2025 – Expected June 2027

BRAC University, Dhaka, Bangladesh

Current CGPA: 3.50/4.00

Bachelor of Science in Computer Science and Engineering

Completed October 2024

BRAC University, Dhaka, Bangladesh

CGPA: 3.09/4.00

Higher Secondary School Certificate

Completed

Cantonment English School & College, Chattogram

GPA: 5.00/5.00

RESEARCH & PUBLICATIONS

Thesis: *Image Processing and Deep Learning for Space Telescope Images: An Exploratory Data Analysis Approach* (Unpublished)

- **Evaluating Large Vision Language Models on Bangla Medical Visual Question Answering** – Published in Findings of ACL 2026.
- **5-DIALECTS-BN: Unmasking the Impact of Transliteration on Bangla Dialectal LLMs** – Accepted at EMNLP 2026 (Main Conference). [First Author]
- **Benchmarking Large Language Models on Bangla Dialect Translation and Dialectal Sentiment Analysis** – BLP 2025 Workshop at IJCNLP-AACL. [First Author; Best Long Paper Award]
- **Linear Probing of Pre-trained Transformer-based Models for Bangla Hate Speech Detection** – BLP 2025 Workshop at IJCNLP-AACL.
- **Exploring Large Language Model and Machine Learning Models for Sentiment Analysis of Bengali Natok Reviews** – Published at ICECET 2025, France.

PROJECTS

- **UNBLIND: Vision-Language Model Acuity Benchmark:** A psychophysics-style benchmark and LoRA fine-tuning study investigating visual limitations in vision-language models and comparing interventions under a cost-matched evaluation protocol.
- **StreamPay: Decentralized Payroll Streaming Protocol:** Developed a Solidity-based application for time-based ETH salary vesting, employee withdrawals, cancellation settlements, and protocol fee accounting, with a MetaMask-connected frontend.
- **BlockSim: Blockchain Simulation and Incentive Analysis:** Extended a Python blockchain simulator to investigate miner participation, block intervals, stale rates, transaction ordering, and reward halving through

configurable experiments and visualizations.

- **MCP Sentiment Analysis:** Exposed TextBlob sentiment analysis through a Gradio-based MCP server and developed a conversational tool client using Hugging Face smolagents.
- **IMDb Sentiment Classification:** Fine-tuned a pretrained BERT model with a custom classification head in PyTorch to classify movie reviews as positive or negative.
- **Street Signboard Detection:** Trained and evaluated a YOLOv8-based object detection model to identify shop signboards in street images.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Kotlin
- **Machine Learning & CV:** TensorFlow, PyTorch, YOLO, Roboflow, CVAT, Grounded SAM 2, RAG, MCP, AI AGENTS
- **Tools & Frameworks:** FastAPI, Docker, Git, Visual Studio, Android Studio, Autodesk Platform Services (APS), Odoo

STANDARDIZED TESTS

GRE (Jan 2025): 302 (Quant: 155, Verbal: 147, Analytical: 4.0)

IELTS (Dec 2024): 7.5 Overall (Listening: 8.0, Reading: 7.5, Writing: 7.5, Speaking: 7.0)

EXTRACURRICULAR & ACHIEVEMENTS

- **Sergeant**, Bangladesh National Cadet Corps (BNCC) (Jan 2014 – Jan 2018)
- **Ranked 22nd Divisional**, National Earth Olympiad (June 2015)
- **General Member**, BRAC University Robotics Club & Computer Club (Jan 2020 – Jan 2023)